

BOBOC AIR BASE, Romania

WMO: 150001

Lat: 45.210N

Lon: 26.972E

Elev: 103

StdP: 100.09

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.1	-10.8	-16.2	0.9	-11.7	-14.0	1.1	-10.1	19.1	-7.9	15.3	-3.6	4.9	20	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.6	33.5	20.9	32.0	20.4	30.8	19.9	22.6	30.3	21.7	29.1	20.9	28.5	4.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.0	14.9	26.6	19.0	14.0	25.5	18.1	13.2	24.6	66.9	30.3	63.9	29.3	61.0	28.6	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 13.2	10.9	10.0		-16.1	36.0	4.2	1.9	-19.1	37.3	-21.5	38.4	-23.9	39.5	-26.9	40.8	
(5)			DB	-15.8	24.0	4.0	1.0	-18.7	24.7	-21.1	25.3	-23.3	25.9	-26.3	26.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	-1.6	0.4	6.3	11.9	17.3	21.1	23.3	23.7	18.4	11.6	6.7	1.2
	DBStd	9.55	5.07	5.50	4.49	3.54	2.92	2.86	2.47	2.76	3.46	4.03	4.30	4.25
	HDD10.0	1196	360	271	129	18	0	0	0	0	30	117	272	
	HDD18.3	2884	618	503	374	195	54	8	1	42	209	349	530	
	CDD10.0	1839	0	1	13	76	227	334	412	424	253	81	18	0
	CDD18.3	485	0	0	0	3	22	92	154	167	45	2	0	0
	CDH23.3	5041	0	0	0	57	249	871	1520	1836	482	26	1	0
	CDH26.7	1868	0	0	0	11	34	274	604	806	138	1	0	0
(14) Wind	WSAvg	4.6	4.9	5.3	5.5	5.1	4.6	4.3	4.1	4.3	4.5	4.4	4.2	4.5
(15) Precipitation	PrecAvg	537	25	19	28	42	67	77	67	52	50	45	32	33
	PrecMax	763	60	56	61	110	195	117	170	129	117	131	76	89
	PrecMin	342	2	2	0	17	19	23	7	10	4	3	1	1
	PrecStd	112	16	15	16	21	41	25	42	34	30	30	22	24
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	15.8	21.9	28.2	29.2	33.2	35.2	36.8	32.1	26.1	21.1	14.1
		MCWB	6.8	11.6	11.9	16.5	17.4	21.9	21.0	21.2	18.6	15.8	14.1	9.6
	2%	DB	8.2	12.1	19.0	25.0	27.2	31.2	33.2	34.0	30.1	23.8	18.0	11.1
		MCWB	5.7	8.4	10.5	15.5	17.5	21.0	20.9	20.7	17.9	15.8	13.1	7.8
	5%	DB	6.1	9.2	16.6	22.0	26.0	29.9	31.9	32.2	28.1	21.2	15.1	8.9
		MCWB	4.8	6.5	9.6	13.8	17.1	20.5	20.7	20.1	17.2	14.5	11.7	6.3
	10%	DB	4.9	7.1	14.0	19.8	24.2	28.1	30.0	31.0	26.1	18.9	13.1	7.0
		MCWB	3.7	5.3	8.4	12.7	16.3	19.7	20.4	19.6	16.3	13.4	10.5	5.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	11.5	13.0	17.4	19.8	23.6	23.2	23.7	19.4	17.9	15.0	10.7
		MCDB	10.2	15.4	19.6	26.3	25.8	30.9	31.0	30.9	27.6	22.9	19.8	13.0
	2%	WB	5.9	9.0	11.2	15.8	18.7	22.4	22.4	22.6	18.6	16.0	13.5	8.4
		MCDB	6.9	12.0	17.7	23.9	25.2	29.5	29.9	30.9	27.6	21.7	17.1	10.5
	5%	WB	4.8	6.8	10.1	14.4	17.7	21.2	21.6	21.2	17.9	14.9	12.3	6.7
		MCDB	5.9	8.5	15.7	21.3	24.2	28.0	29.0	30.0	26.7	20.7	15.1	7.9
	10%	WB	3.6	5.3	8.7	13.0	16.8	20.3	21.0	20.4	17.0	13.8	10.9	5.3
		MCDB	4.5	6.9	13.4	18.1	23.1	26.6	28.2	29.2	25.1	18.2	13.0	6.7
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	7.3	10.2	11.8	11.6	11.2	11.7	12.6	12.1	10.2	8.0	7.1
		MCDBR	9.0	11.5	15.5	16.9	14.4	13.7	14.2	14.6	15.2	13.8	12.0	10.5
	5% WB	MCWBR	6.9	8.2	9.0	9.0	7.0	6.2	5.4	5.4	6.2	7.5	7.7	7.8
		MCDWR	8.4	10.5	14.0	15.1	12.7	12.1	12.3	13.1	14.2	13.3	10.8	9.4
(40) Clear-Sky Solar Irradiance	taub		0.332	0.362	0.396	0.455	0.429	0.431	0.446	0.460	0.421	0.384	0.357	0.333
		taud	2.332	2.249	2.208	2.063	2.218	2.246	2.211	2.160	2.234	2.320	2.348	2.343
	Ebn at Noon		762	803	826	805	843	840	822	792	792	769	723	716
		Edh at Noon	79	104	124	156	138	135	138	140	120	96	78	72
(44) All-Sky Solar Radiation	RadAvg		1.31	2.13	3.30	4.53	5.71	6.08	6.18	5.57	4.05	2.62	1.44	1.10
	RadStd		0.17	0.27	0.34	0.47	0.44	0.44	0.44	0.36	0.34	0.30	0.15	0.14

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (59 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	-137	+70	+38

Nomenclature: See separate page